

Review: Operations with rational numbers



1

a 2.1

b -2.1

c 32.31

d -7.71

e -4.8

f -124.2

g 7.2

h 3.1

i -54.2

j 18.9

k $-\frac{3}{4}$

l $85\frac{2}{5}$

2 No

3 \$72.18

4 $\frac{12}{35}$

5 $\frac{15}{28}$

Additional questions

6

a $\frac{1}{3}$

b $\frac{1}{3}$

c $\frac{3}{10}$

d $\frac{3}{5}$

e $\frac{9}{4}$

f $\frac{4}{3}$

g $\frac{13}{25}$

h $\frac{4}{11}$

i $\frac{2}{5}$

j $\frac{16}{23}$

k $\frac{4}{5}$

l $\frac{3}{10}$

m $\frac{1}{5}$

n $\frac{1}{8}$

o $\frac{1}{11}$

p $\frac{5}{9}$

7

a $1\frac{4}{7}$

b $2\frac{2}{5}$

c $4\frac{1}{2}$

d $1\frac{2}{5}$

8

$$\begin{array}{l} \text{a} \quad 1\frac{4}{9} \\ \text{e} \quad \frac{7}{30} \\ \text{i} \quad \frac{24}{35} \\ \text{m} \quad -\frac{2}{7} \\ \text{q} \quad \frac{36}{49} \\ \text{u} \quad -1\frac{1}{15} \end{array}$$

$$\begin{array}{l} \text{b} \quad \frac{7}{10} \\ \text{f} \quad \frac{5}{8} \\ \text{j} \quad 17\frac{1}{2} \\ \text{n} \quad \frac{16}{81} \\ \text{r} \quad \frac{3}{20} \\ \text{v} \quad \frac{11}{15} \end{array}$$



$$\begin{array}{l} \text{c} \quad 1\frac{17}{24} \\ \text{g} \quad 2\frac{1}{18} \\ \text{k} \quad 1 \\ \text{o} \quad \frac{6}{35} \\ \text{s} \quad \frac{1}{2} \\ \text{w} \quad \frac{19}{35} \end{array}$$

$$\begin{array}{l} \text{d} \quad 2\frac{13}{15} \\ \text{h} \quad \frac{1}{7} \\ \text{l} \quad \frac{2}{81} \\ \text{p} \quad \frac{81}{100} \\ \text{t} \quad \frac{7}{25} \end{array}$$

9

$$\begin{array}{l} \text{a} \quad 6\frac{10}{11} \\ \text{e} \quad 7\frac{4}{5} \\ \text{i} \quad 1\frac{2}{11} \\ \text{m} \quad 1\frac{1}{4} \\ \text{q} \quad 1 \\ \text{u} \quad 1\frac{13}{15} \end{array}$$

$$\begin{array}{l} \text{b} \quad 8\frac{2}{15} \\ \text{f} \quad 10\frac{3}{4} \\ \text{j} \quad 1\frac{3}{28} \\ \text{n} \quad 5\frac{1}{2} \\ \text{r} \quad 3\frac{3}{5} \\ \text{v} \quad 1\frac{17}{28} \end{array}$$

$$\begin{array}{l} \text{c} \quad 8\frac{1}{4} \\ \text{g} \quad 15\frac{3}{8} \\ \text{k} \quad 2\frac{2}{5} \\ \text{o} \quad 4\frac{2}{3} \\ \text{s} \quad \frac{7}{9} \\ \text{w} \quad \frac{18}{49} \end{array}$$

$$\begin{array}{l} \text{d} \quad 6\frac{2}{5} \\ \text{h} \quad 3\frac{2}{9} \\ \text{l} \quad 6\frac{1}{8} \\ \text{p} \quad 6\frac{2}{7} \\ \text{t} \quad 7\frac{1}{2} \\ \text{x} \quad 1\frac{1}{3} \end{array}$$

10

$$\begin{array}{l} \text{a} \quad \frac{272}{175} \\ \text{e} \quad \frac{5}{72} \\ \text{i} \quad \frac{7}{10} \\ \text{m} \quad \frac{26}{3} \end{array}$$

$$\begin{array}{l} \text{b} \quad \frac{7}{8} \\ \text{f} \quad \frac{16}{49} \\ \text{j} \quad \frac{3}{8} \\ \text{n} \quad \frac{11}{20} \end{array}$$

$$\begin{array}{l} \text{c} \quad \frac{2}{35} \\ \text{g} \quad \frac{49}{37} \\ \text{k} \quad \frac{10}{21} \end{array}$$

$$\begin{array}{l} \text{d} \quad \frac{1}{72} \\ \text{h} \quad \frac{63}{64} \\ \text{l} \quad \frac{7}{10} \end{array}$$

11

$$\begin{array}{l} \text{a} \quad 46.1 \\ \text{e} \quad 2.46 \\ \text{i} \quad 3.84 \\ \text{m} \quad 0.89 \\ \text{q} \quad 4 \\ \text{u} \quad 1.70 \end{array}$$

$$\begin{array}{l} \text{b} \quad 13.658 \\ \text{f} \quad 500 \\ \text{j} \quad 8.46 \\ \text{n} \quad 13 \\ \text{r} \quad 600 \\ \text{v} \quad 255.71 \end{array}$$

$$\begin{array}{l} \text{c} \quad 47 \\ \text{g} \quad 2.355 \\ \text{k} \quad 38.4 \\ \text{o} \quad 4.46 \\ \text{s} \quad 9 \\ \text{w} \quad 5.35 \end{array}$$

$$\begin{array}{l} \text{d} \quad 15.35 \\ \text{h} \quad 8.8 \\ \text{l} \quad 6.1908 \\ \text{p} \quad 80 \\ \text{t} \quad 56.24 \end{array}$$



12

a \$34.77

b \$64.47

c \$254.90

d \$32.46

13

a \$27.80

b \$64.85

c \$34.90

d \$57.55

e \$227.00

f \$634.65

g \$1025.00

h \$21.95

14

a
i 10

ii 9.46

iii 0.54

iv No solution
provided.

b
i 2

ii 1.78

iii 0.22

iv No solution
provided.

c
i 3

ii 2.61

iii 0.39

iv No solution
provided.

15

a
i 3

ii 4.41

b
i 3

ii 3.01

16 52.781 19

17

a A

b 29.5 g

18

a C

b \$60.98

19

a Luigi

b $\frac{13}{14}$

c $\frac{1}{14}$

20

a $\frac{13}{40}$

b $\frac{27}{40}$

21 a $\frac{3}{5}$

b 24 girls



22 a $\frac{1}{12}$

b 96 dice

23 a $\frac{19}{90}$

b $\frac{71}{90}$

c 213 people

24 a 270 pages

b 90 pages

25 a 127 L

b 508 L

26 $\frac{5}{6}$ hour

27 $\frac{19}{35}$ mi

28 $\frac{5}{28}$ ft

29 \$21.95

30 \$1.80

31 \$35.05

32 \$79.61

33 \$142.77

34 \$746.82

35 \$34.06

36 \$8904



37 3000 hot chocolates

38 41 m^2

39 $\frac{4}{27} \text{ lb}$

40 $\frac{5}{18} \text{ lb}$

41 0.4 lb

42 36 pieces

43 41 pieces

44 110 bottles

45 420 mL

46 53 computers

47 \$365.57

48
a \$242

b \$434

49
a 477 cm

b 4.77 m

50 401.6 m

51 16.6 km

52 0.65 seconds

53

a 180 seconds

b 40 pictures



54

a $\frac{1}{4}$

b $\frac{3}{32}$ ft

55 $14\frac{2}{5}$ m

56 600 cars

57 300 trucks

58 45.1°F

59 68.44°F

60

a -5.15 ft

b 1.2 ft/s